Idea generation & development

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe the key stages of brainstorming
- ☐ describe the key stages of morphological analysis
- ☐ explain why idea-generation techniques are used
- ☐ describe the key stages of analogy
- ☐ describe how lifestyle and mood boards are used
- ☐ explain exploration, synthesis and refinement of ideas

I DO — Worked example

Read how a full-mark answer is built

A design team needs ideas for a new bike light.

1 Describe three key stages of brainstorming. (3 marks)

Full-mark answer each ✓ is one mark

The problem is explained clearly to a small group so everyone knows the purpose ✓. The group suggests as many ideas as possible within a time limit, and every idea is recorded ✓. No idea is criticised; members build on each other's ideas before the best are selected ✓.

Why marks are not awarded

"You think of lots of ideas and write them down."

0–1 mark: "think of lots of ideas" was given 0; only recording ideas could score.

WE DO — Guided practice

Use the starters and word bank to help you

2 Describe two idea-generation techniques. (4 marks)

Sentence starters

Brainstorming: …
Morphological analysis: …
Analogy: …
Mood board: …

Word bank

problem · Mood board · simply · stated · technique · Analogy · Morphological analysis · brainstorming

Tip: 4 marks = 4 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Combining the best features of several initial ideas is called...
☐ Refinement ☐ Synthesis ☐ Evaluation ☐ Analysis

b) Which is a rule of brainstorming?
☐ Only feasible ideas are recorded ☐ Ideas are judged as they appear ☐ Every idea is recorded without criticism ☐ It must be done alone

c) Nike basing a trainer's cushioning on Formula One suspension is an example of...
☐ Mood board ☐ Analogy ☐ Morphological analysis ☐ Synthesis

d) How does morphological analysis improve the breadth of ideas?
☐ It removes weak ideas ☐ It creates many unique combinations of variables ☐ It tests ideas with users ☐ It records ideas as sketches

4 Key terms: write the letter of the matching description next to each term.

1. **Morphological analysis**	____	**A** Collage of images on a mood/theme to inspire ideas and set an aesthetic direction.
2. **Brainstorming rule**	____	**B** Taking an idea from one context and applying it in another (e.g. F1 suspension → trainer cushioning).
3. **Analogy**	____	**C** A matrix of design variables and options; combine one option from each to generate many ideas.
4. **Lifestyle board**	____	**D** Record every idea and don't criticise; build on each other's ideas within a time limit.
5. **Mood board**	____	**E** Images showing the lifestyle of the typical user, to understand the target market.

5 Explain what is meant by ‘Synthesis’. (2 marks)

6 Explain what is meant by ‘Refinement’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?